Proposal Outline for Coding the Shape of Nuclei

Name: Ripley, George
Project Name: Coding the Shape of Nuclei

Project Supplies Needed:

- Laptop
- Visual Studio Code (Or really any IDE that is Python-compatible)
- Python 3.13-3.14
- Operating System (Linux, Mac, Windows)
- Git Repository
- 2D Graphical Engine (PhysEng or Pymunk)
- Possible Screen for people to view it at the fair?

Estimated cost, list every item and cost as well as total:

Item	Link	Quantity	Price	Total Price
(My own) Laptop	-	1	$0	$0
Visual Studio Code	https://code.visualstudio.com/	1	$0	$0
Python 3.14	https://www.python.org/downloads/release/python-3147/	1	$0	$0
Operating System (Linux, Mac, Windows.)	https://linuxmint.com/download.php	1	$0	$0
Git Repository	https://github.com/ripRoot/Shape-of-Nuclei/tree/main	1	$0	$0
PhysEng/Pymunk ?	https://github.com/levi2234/PhysEng https://www.pymunk.org/en/latest/	1	$0	$0

Numpy	https://numpy.org/	1	$0	$0

How will you assemble it?
I will create a folder for the project to be built in. In that folder, there will be a virtual environment that will have the different python libraries, PhysEng and Numpy. We will collaborate to build a program using the below logic (Rudimentary → Advanced.)

- Having a marble roll in at a randomized position to then be marked to a correlating position on the perimeter of a shape, then having the marble (drift, roll, drag?) towards the position. Then, have it stop and fade to half of its opacity to start to mark the outline of the shape. It will be on a dark background, and the marbles would be white.

Possible challenges
With the initial rudimentary description of the logic for the program, we might encounter horrible time complexity. But that can be fixed with time.
How will we show it to the public at their fair? On a larger screen/monitor? (I have a large-ish monitor I can bring and hook up when the time comes, but transporting it will be difficult for me.)